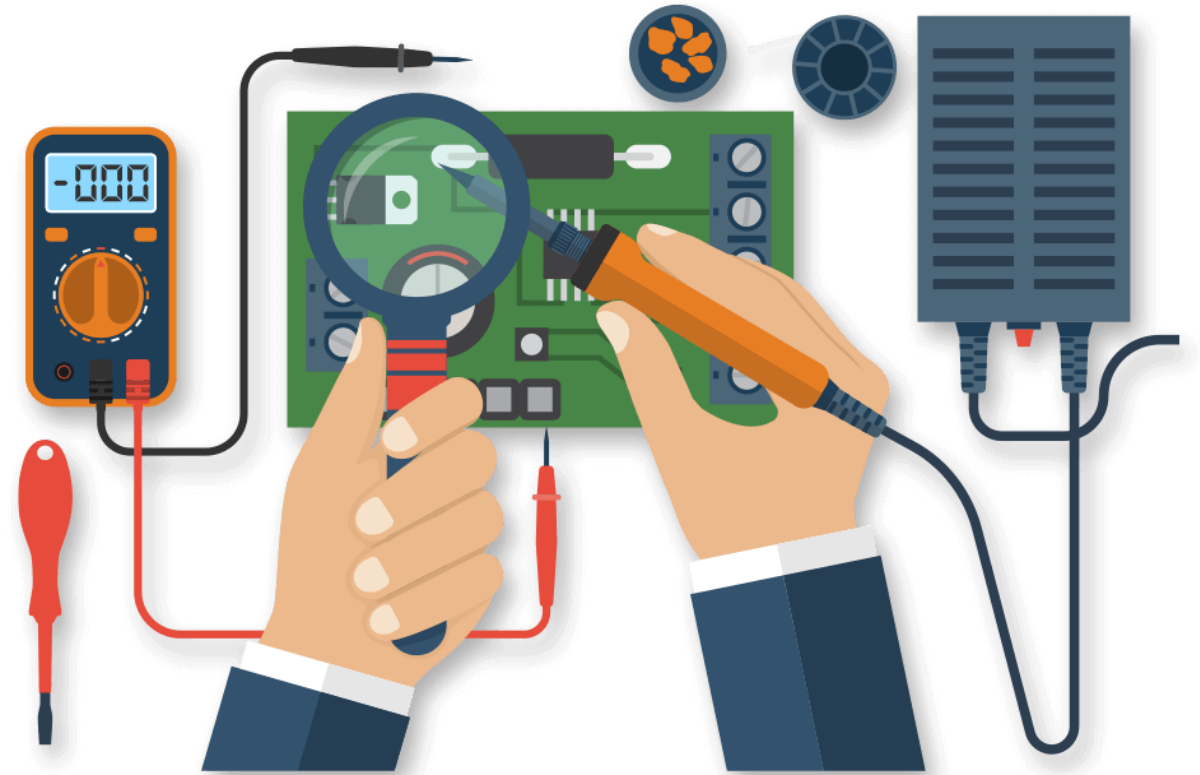


Computer System Architecture

Day 2: Voltage, Current, and Resistance



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Outline

- Electricity
- Voltage
- Current
- Resistance



Electricity

Electricity is like a water hose

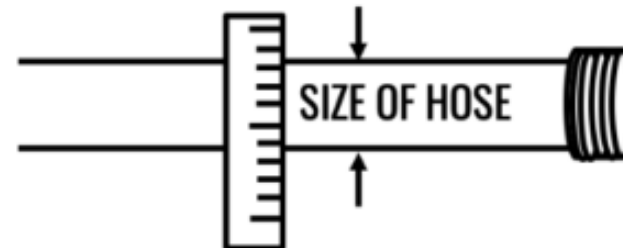
Voltage

Volts (V)



Current

Amps (A or I)

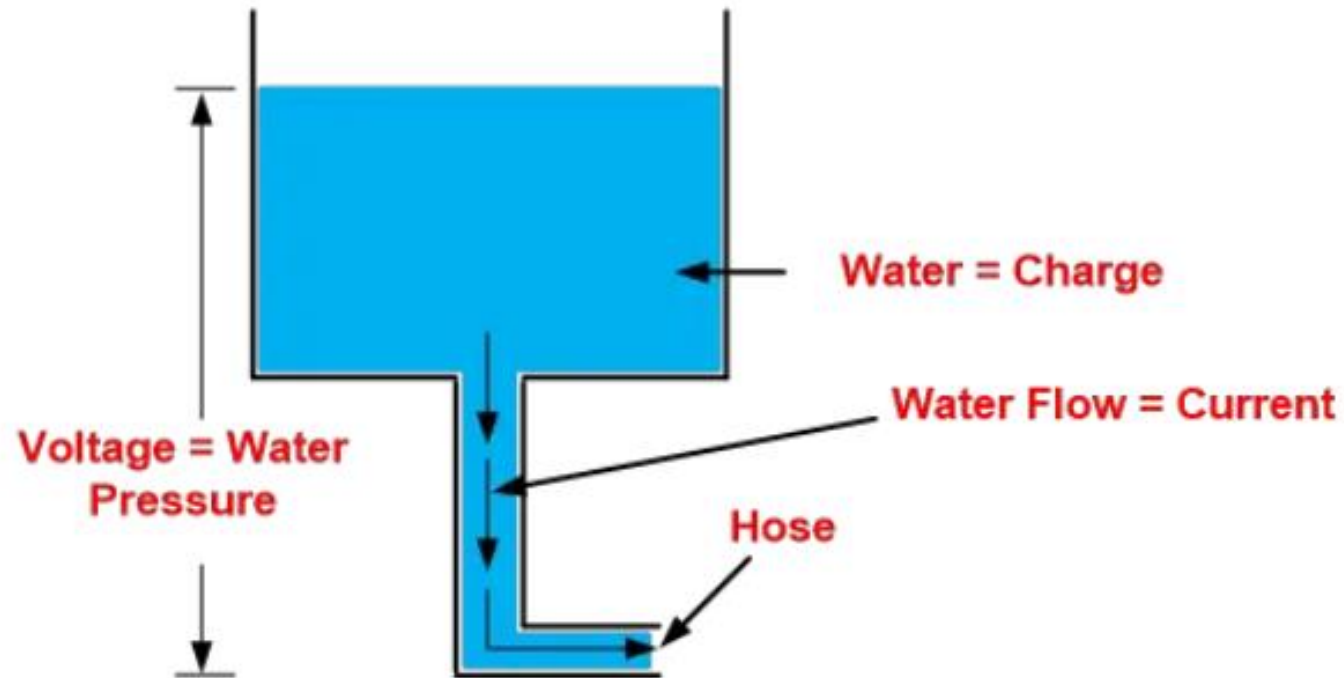


Resistance

Ohms (R or Ω)

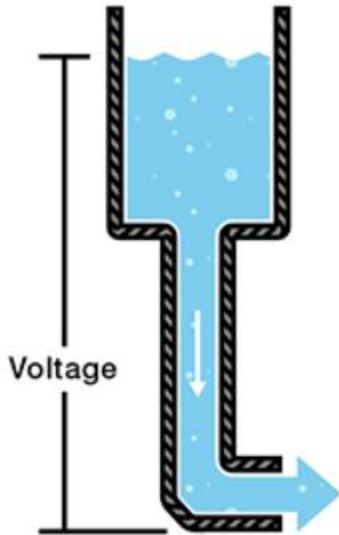


Electricity



Hydraulic Analogy 2

Voltage



Water = Charge

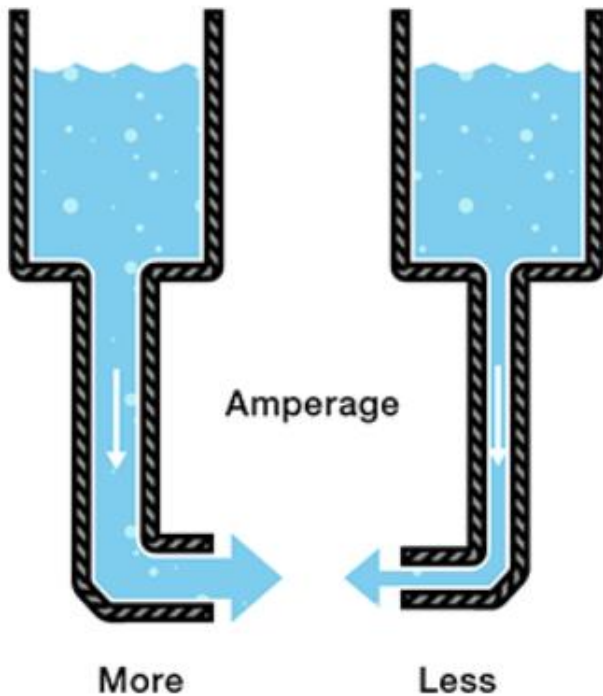
Pressure = Voltage (measured in **Volts**)

Flow = Current

Hose Width = Resistance

Voltage is the difference in charge between two points.

Current



Water = Charge

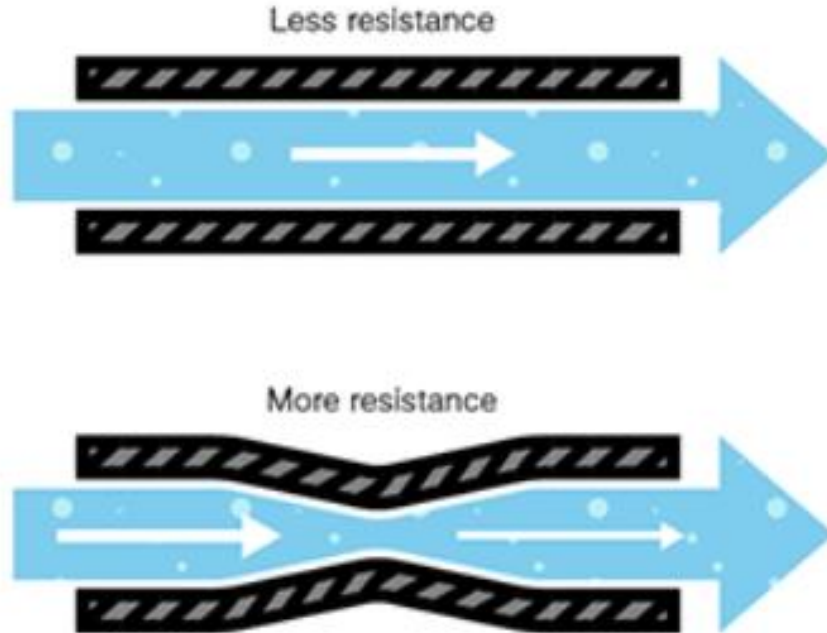
Pressure = Voltage

Flow = Current (measured in **Amperes**)

Hose Width = Resistance

Current is the rate at which charge is flowing

Resistance



Water = Charge
Pressure = Voltage
Flow = Current

Hose Width = Resistance (measured in **Ohms**)

Resistance is a material's tendency to resist the flow of charge (current).

Thank you!



Questions

